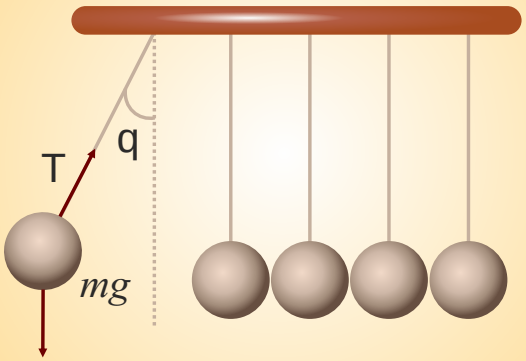
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
path to success

PHYSICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Newton's Law of Motion & Friction

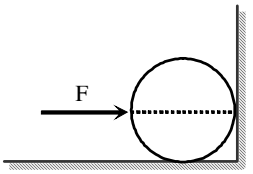
(English Medium)

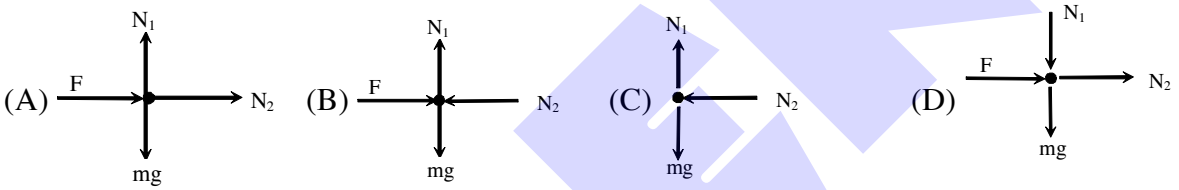
EXERCISE (O)

SINGLE CORRECT TYPE QUESTIONS

Newton's Laws of Motion:

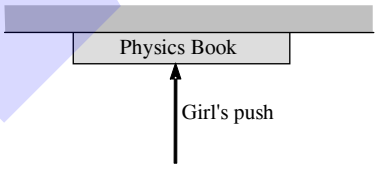
1. A ball of mass m kept at the corner as shown in the figure, is acted by a horizontal force F. The correct free body diagram of ball is





NL0036

2. A girl pushes her physics book up against the horizontal ceiling of her room as shown in the figure. The book weighs 20 N and she pushes upwards with a force of 25 N. The choices below list the magnitudes of the contact force F_{CB} between the ceiling and the book, and F_{BH} between the book and her hand. Select the correct pair.



(A) F_{CB} = 20 N and F_{BH} = 25 N (B) F_{CB} = 25 N and F_{BH} = 45 N

(C) F_{CB} = 5 N and F_{BH} = 25 N (D) F_{CB} = 5 N and F_{BH} = 45 N **NL0042**

3. Two astronauts A and B connected with a rope stay stationary in free space relative to their spaceship. Mass of A is more than that of B and the rope is straight. Astronaut A starts pulling the rope but astronaut B does not. If you were the third astronaut in the spaceship, what do you observe?

(A) Astronaut B accelerates towards A and A remains stationery.

(B) Both accelerate towards each other with equal accelerations of equal modulus.

(C) Both accelerate towards each other but acceleration of B is greater than that of A.

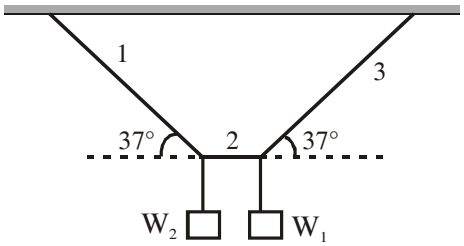
(D) Both accelerate towards each other but acceleration of B is smaller than that of A.

NL0043

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture \Phy\Module\1-U & D, Basic Maths & Vector, NLM\K-1 D,2D\Eng\Eng\04_NLM & Friction\01_Engp65

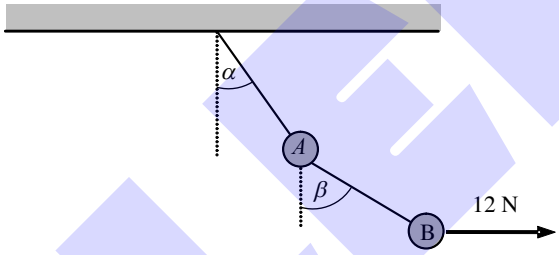
4. Under what condition(s) will an object be in equilibrium ?
(A) Only if it is at rest
(B) Only if it is moving with constant velocity
(C) Only if it is moving with constant acceleration
(D) If it is either at rest or moving with constant velocity **NL0038**

5. In a given figure system is in equilibrium. If $W_1 = 300$ N. Then W_2 is approximately equal to



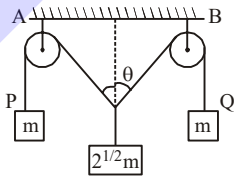
(A) 500 N (B) 400 N (C) 670 N (D) 300 N **NL0040**

6. Two balls A and B weighing 7 N and 9 N are connected by a light cord. The system is suspended from a fixed support by connecting the ball A with another light cord. The ball B is pulled aside by a horizontal force 12 N and equilibrium is established. Angles α and β respectively are



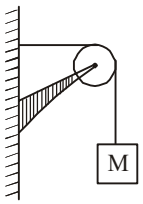
(A) 30° and 60° (B) 60° and 30° (C) 37° and 53° (D) 53° and 37°
NL0041

7. The pulleys and strings shown in the figure are smooth and of negligible mass. For the system to remain in equilibrium, the angle θ should be **[JEE (Scr) 2001]**



(A) 0° (B) 30° (C) 45° (D) 60° **NL0048**

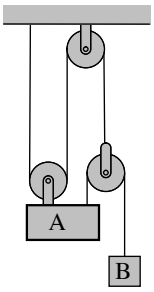
8. A string of negligible mass going over a clamped pulley of mass *m* supports a block of mass *M* as shown in the figure. The force on the pulley by the clamp is given **[JEE (Scr) 2001]**



(A) $\sqrt{2}\,Mg$ (B) $\sqrt{2}\,mg$ (C) $\sqrt{(M+m)^2+m^2}\,g$ (D) $\sqrt{(M+m)^2+M^2}\,g$
NL0056

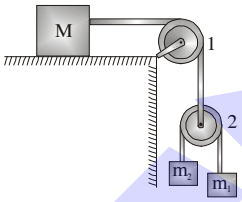
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\11-U & D, Basic Maths & Vector, NUM, K-11D,2D\Eng\Eng\04_NUM & Friction\01_Eng.p65

9. In arrangement shown the block A of mass 15 kg is supported in equilibrium by the block B. Mass of the block B is closest to



(A) 2 kg (B) 3 kg (C) 4 kg (D) 5 kg **NL0053**

10. In the arrangement shown in figure $m_1 = 1kg$, $m_2 = 2kg$. Pulleys are massless and strings are light. For what value of M the mass m_1 moves with constant velocity (Neglect friction)



(A) 6 kg (B) 4 kg (C) 8 kg (D) 10 kg

NL0095

11. A flexible chain of weight W hangs between two fixed points A & B which are at the same horizontal level. The inclination of the chain with the horizontal at both the points of support is θ. What is the tension of the chain at the mid point?

(A) $\frac{W}{2}.\,cosec\,\theta$ (B) $\frac{W}{2}.\,tan\,\theta$ (C) $\frac{W}{2}cot\,\theta$ (D) none

NL0094

12. Four blocks of same mass connected by cords are pulled by force F on a smooth horizontal surface, as in figure. The tension T_1, T_2 and T_3 will be



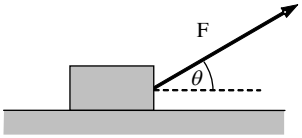
(A) $T_1 = F/4$, $T_2 = 3F/2$, $T_3 = F/4$ (B) $T_1 = F/4$, $T_2 = F/2$, $T_3 = F/2$
(C) $T_1 = 3F/4$, $T_2 = F/2$, $T_3 = F/4$ (D) $T_1 = 3F/4$, $T_2 = F/2$, $T_3 = F/2$ **NL0039**

13. A man is standing on a weighing machine with a block in his hand. The machine records w. When he takes the block upwards with some acceleration the machine records w_1. When he takes the block down with some acceleration, the machine records w_2.Then choose correct option

(A) $w_1 = w = w_2$ (B) $w_1 < w < w_2$ (C) $w_2 < w < w_1$ (D) $w_2 = w_1 > w$

NL0045

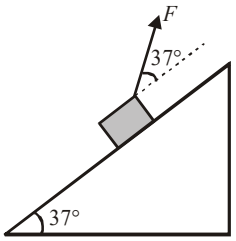
14. A block is being pulled by a force F on a long frictionless level floor. Magnitude of the force is gradually increases from zero until the block lifts off the floor. Immediately before the block leaves the floor, its acceleration is



(A) gcosθ (B) gcotθ (C) gsinθ (D) g tanθ **NL0046**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture \Phy\Module\1-U & D, Basic Maths & Vector, NUM\K-1 D,2D\Eng\Eng\04_NUM & Friction\01_Engp65

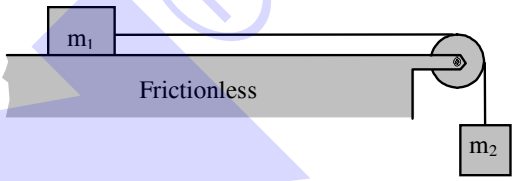
15. A block resting on a smooth inclined plane is acted upon by a force F as shown. If mass of block is 2 kg and F = 20 N and sin 37°= 3/5, the acceleration of block is



(A) 2 m/s^2 (B) 6 m/s^2 (C) 8 m/s^2 (D) zero

NL0049

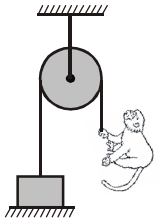
16. In the arrangement shown, the blocks of unequal masses are held at rest. When released, acceleration of the blocks is



(A) g/2.
(B) g.
(C) a value between zero and g.
(D) a value that could be greater than g.

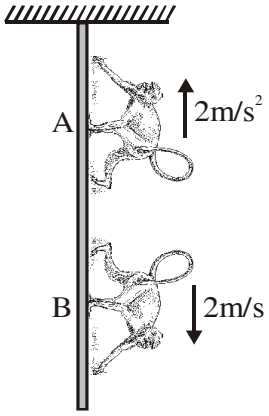
NL0050

17. A monkey weighing 10 kg is climbing up a light rope and frictionless pulley attached to 15 kg mass at other end as in figure. In order to raise the 15 kg mass off the ground the monkey must climb-up



(A) with constant acceleration g/3.
(B) with an acceleration greater than g/2.
(C) with an acceleration greater than g/4.
(D) It is not possible because weight of monkey is lesser than the block. **NL0051**

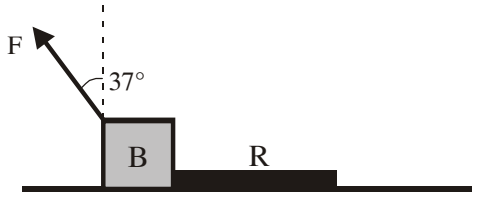
18. Two monkeys of masses 10 kg and 8 kg are moving along a vertical light rope, the former climbing up with an acceleration of 2 m/s^2, while the latter coming down with a uniform velocity of 2 m/s. Find tension in the rope at the fixed support.



(A) 180 N (B) 200 N (C) 80 N (D) 216 N **NL0085**

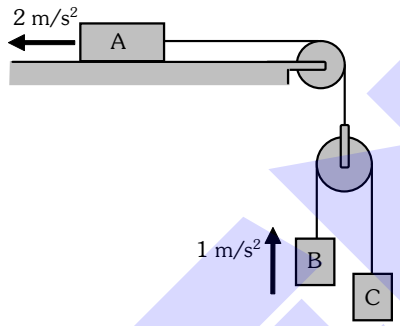
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\11-U & D, Basic Maths & Vector, NLM, K-11 D, 2D\Eng\Eng\04_NLM & Friction\01_Eng.p65

19. A block B is tied to one end of a uniform rope R as shown. The mass of block is 2 kg and that of rope is 1 kg. A force F = 15 N is applied at angle 37° with vertical. The tension at the mid-point of rope is



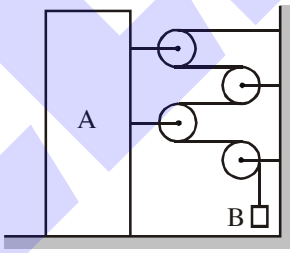
(A) 1.5 N (B) 2 N (C) 3N (D) 4.5 N **NL0047**

20. In the setup shown, find acceleration of the block C.



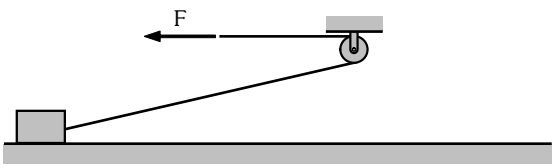
(A) 3 m/s² ↑ (B) 3 m/s² ↓ (C) 5 m/s² ↑ (D) 5 m/s² ↓ **NL0063**

21. Block A is moving away from the wall at a speed v and acceleration a.



(A) Velocity of B is v with respect to A. (B) Acceleration of B is a with respect to A.
(C) Acceleration of B is 4a with respect to A. (D) Acceleration of B is √17a with respect to A.
NL0062

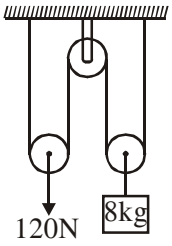
22. A heavy cart is pulled by a constant force F along a horizontal track with the help of a rope that passes over a fixed pulley, as shown in the figure. Assume the tension in the rope and the frictional forces on the cart remain constant and consider motion of the cart until it reaches vertically below the pulley. As the cart moves to the right, its acceleration



(A) decreases. (B) increases. (C) remains constant. (D) is zero **NL0052**

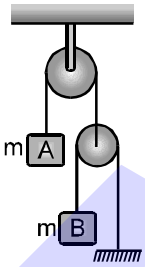
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\1-U & D, Basic Maths & Vector, NLM\K-1 D,2D\Eng\Eng\04_NLM & Friction\01_Engp05

23. If the string is pulled down with a force of 120 N as shown in the figure, then the acceleration of 8 kg block would be



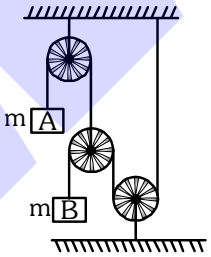
(A) 10 m/s^2 (B) 5 m/s^2 (C) 0 m/s^2 (D) 4 m/s^2 **NL0143**

24. In the figure shown neglecting friction and mass of pulleys, what is the acceleration of mass B?



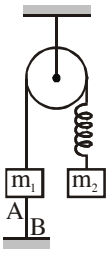
(A) $\frac{g}{3}$ (B) $\frac{5g}{2}$ (C) g (D) $\frac{2g}{5}$ **NL0144**

25. In the arrangement shown, the pulley and the strings are ideal. The acceleration of block B is–



(A) g/5 (B) g/2 (C) 2g/5 (D) 2g/3 **NL0145**

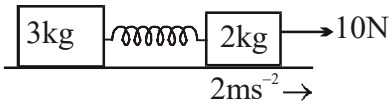
26. In a given figure two masses m_1 & m_2 ($m_2 > m_1$) are at rest in equilibrium position. Find the tension in string AB



(A) m_1g (B) m_2g (C) $(m_1+m_2)g$ (D) $(m_2-m_1)g$ **NL0057**

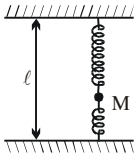
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\11-U & D, Basic Maths & Vector, NLM,K-11D,2D\Eng\Eng\04_NLM & Friction\01_Eng.p65

27. Find the acceleration of 3 kg mass when acceleration of 2 kg mass is 2 ms^{-2} as shown in figure.



(A) 3 ms^{-2} (B) 2 ms^{-2} (C) 0.5 ms^{-2} (D) zero **NL0059**

28. A small ball of mass M is held in equilibrium with two identical springs as shown in the figure . Force constant of each spring is k and relaxed length of each spring is $\ell/2$. What is distance between the ball and roof?



(A) $\frac{\ell}{2}+\frac{Mg}{k}$ (B) $\frac{\ell}{2}-\frac{Mg}{k}$ (C) $\frac{\ell}{2}+\frac{Mg}{2k}$ (D) $\frac{\ell}{2}-\frac{Mg}{2k}$

NL0060

29. An elastic spring of relaxed length ℓ_o and force constant k is cut into two parts of lengths ℓ_1 and ℓ_2. The force constants of these parts are respectively

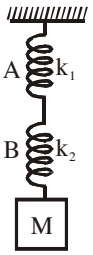
(A) $\frac{k\ell_o}{\ell_1}$ and $\frac{k\ell_o}{\ell_2}$ (B) $\frac{k\ell_1}{\ell_0}$ and $\frac{k\ell_2}{\ell_o}$ (C) $\frac{k\ell_0}{\ell_2}$ and $\frac{k\ell_0}{\ell_1}$ (D) $\frac{k\ell_2}{\ell_0}$ and $\frac{k\ell_1}{\ell_0}$

NL0061

30. A load attached to the end of a spring and in equilibrium produces 9 cm extension of spring. If the spring is cut into three equal parts and one end of each is fixed at 'O' and other ends are attached to the same load, the extension in cm of the combination in equilibrium now is: **NL0146**

(A) 1 (B) 3 (C) 6 (D) 9

31. A mass M is suspended by two springs A and B of force constant k_1 & k_2 respectively as shown in the diagram. The total stretch of springs in equilibrium is

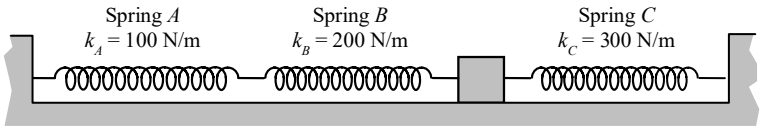


(A) $\frac{Mg}{k_1+k_2}$ (B) $\frac{2Mg}{(k_1+k_2)}$

(C) $\frac{Mg}{2(k_1+k_2)}$ (D) $\frac{Mg(k_1+k_2)}{k_1k_2}$

NL0147

32. In the setup shown, the block is in equilibrium and all the three springs are relaxed. Spring A is attached to the left wall at one end and at the other end is connected to spring B, which is connected to the block. One end of the spring C is attached to the right wall and the other end to the block. The block rests on the frictionless ground. Now the block is shifted towards the right wall extending the springs A and B and compressing the spring C. If magnitudes of the forces developed in these springs are F_A, F_B and F_C, which of the following is correct expression?

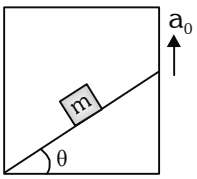


(A) $F_A < F_B < F_C$ (B) $F_A > F_B < F_C$ (C) $F_A = F_B < F_C$ (D) $F_A = F_B > F_C$

NL0148

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture \Phy\Module\1-U & D, Basic Maths & Vector, NUM, K-1 D,2D\Eng\Eng\04_NUM & Friction\01_Engp65

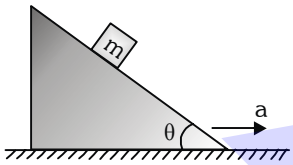
33. In the given figure box is moving vertically upward with acceleration a_0. (all surfaces are smooth), normal force between block and inclined is:



(A) mg cos θ (B) $m(g + a_0)\cos\theta$ (C) $m(g - a_0)\cos\theta$ (D) $\frac{m}{2}(g + a_0)\sin\theta$

NL0149

34. In the given figure if acceleration of block with respect to wedge is zero. Then find the value of 'a' (Neglect Friction)



(A) g cotθ (B) g sinθ (C) g tanθ (D) g cosθ

NL0150

35. In above question find normal force between block and wedge.
(A) mg cosθ (B) mg sec θ (C) mg tan θ (D) mg cosec θ

NL0151

36. A piece of wire is bent in the shape of a parabola $y = kx^2$ (y-axis vertical) with a bead of mass m on it. The bead can slide on the wire without friction. It stays at the lowest point of the parabola when the wire is at rest. The wire is now accelerated parallel to the x-axis with a constant acceleration a. The distance of the new equilibrium position of the bead, where the bead can stay at rest with respect to the wire, from the y-axis is **[IIT-JEE-2009]**

(A) $\frac{a}{gk}$ (B) $\frac{a}{2gk}$ (C) $\frac{2a}{gk}$ (D) $\frac{a}{4gk}$ **NL0116**

Friction:

37. The coefficient of friction between a body and ground is $1/\sqrt{3}$ then (assuming only external horizontal force can act on the body) :

(A) The angle between contact force and normal force can vary from 60° to 90°.

(B) The angle between contact force and normal force can vary from 0° to 30°.

(C) The angle between contact force and normal force can vary from 0° to 60°.

(D) The angle between contact force and normal force can be vary from 30° to 90°.

NL0152

38. A body of mass M is kept on a rough horizontal surface (friction cofficient = μ). A person is trying to pull the body by applying a horizontal force but the body is not moving. The force by the surface on A is F where

(A) F = Mg (B) $F = \mu Mg$

(C) $Mg \le F \le Mg\sqrt{1+\mu^2}$ (D) $Mg \ge F \ge Mg\sqrt{1-\mu^2}$

NL0153

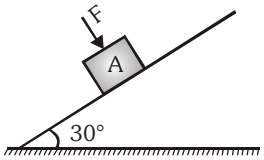
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\11-U & D, Basic Maths & Vector, NUM, K-11 D,2D\Eng\Eng\04_NUM & Friction\01_Eng.p65

39. Starting from rest a body slides down a 45° inclined plane in twice the time it takes to slide down the same distance in the absence of friction. The co-efficient of friction between the body and the inclined plane is:

(A) 0.75 (B) 0.33 (C) 0.25 (D) 0.80

NL0154

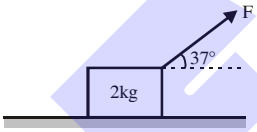
40. A block of mass m = 2 kg is resting on a rough inclined plane of inclination 30° as shown in figure. The coefficient of friction between the block and the plane is μ = 0.5. What minimum force F should be applied perpendicular to the plane on the block, so that block does not slip on the plane ? ($g=10 m/s^2$)



(A) zero (B) 6.24 N (C) 2.68 N (D) 4.34 N

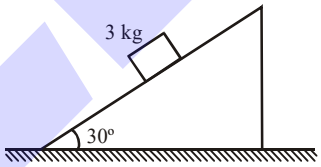
NL0155

41. A block of mass 2 kg is kept on a rough horizontal floor and pulled with a force F. If the coefficient of friction is 0.5. then the minimum force required to move the block is :-



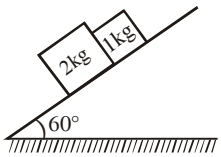
(A) 10 N (B) $\frac{100}{11}$ N (C) $\frac{100}{8}$ N (D) 20 N **NL0064**

42. A block of mass 3 kg is at rest on a rough inclined plane as shown in the figure. The magnitude of net force exerted by the surface on the block will be ($g = 10 m/s^2$)



(A) 26 N (B) 19.5 N (C) 10 N (D) 30 N **NL0068**

43. In the figure shown if friction coefficient of block 1kg and 2kg with inclined plane is μ_1=0.5 and μ_2 = 0.4 respectively, then



(A) both block will move together.

(B) both block will move separately.

(C) there is a non zero contact force between two blocks.

(D) none of these

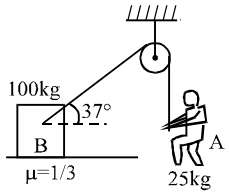
NL0070

44. A block is pushed with some velocity up a rough inclined plane. It stops after ascending few meters and then reverses its direction and returns back to point from where it started. If angle of inclination is 37° and the time to climb up is half of the time to return back then coefficient of friction is

(A) $\frac{9}{20}$ (B) $\frac{7}{5}$ (C) $\frac{7}{12}$ (D) $\frac{5}{7}$ **NL0071**

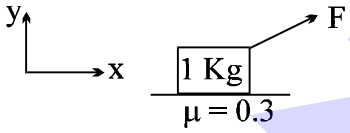
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture \Phy\Module\1-U & D, Basic Maths & Vector, NLM\K-1 D,2D\Eng\Eng\04_NLM & Friction\01_Engp05

45. Block *B* of mass 100 kg rests on a rough surface of friction coefficient $\mu = 1/3$. A rope is tied to block *B* as shown in figure. The maximum acceleration with which boy *A* of 25 kg can climbs on rope without making block move is :



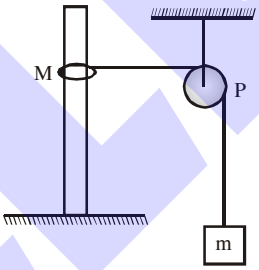
(A) $\frac{4g}{3}$ (B) $\frac{g}{3}$ (C) $\frac{g}{2}$ (D) $\frac{3g}{4}$ **NL0067**

46. A force $\vec{F} = \hat{i} + 4\hat{j}$ acts on block shown. The force of friction acting on the block is :



(A) $-\hat{i}$ (B) $-1.8\hat{i}$ (C) $-2.4\hat{i}$ (D) $-3\hat{i}$ **NL0066**

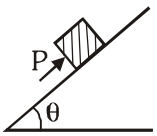
47. In the figure shown a ring of mass *M* and a block of mass *m* are in equilibrium. The string is light and pulley *P* does not offer any friction and coefficient of friction between pole and *M* is μ. The frictional force offered by the pole on *M* is

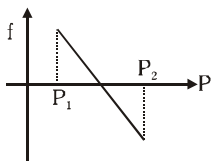


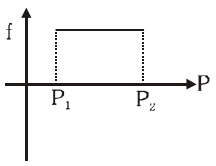
(A) *Mg* directed up (B) μ*mg* directed up
(C) (*M* –*m*)g directed down (D) μ*mg* directed down **NL0065**

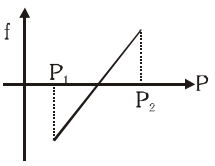
48. A block of mass m is on an inclined plane of angle θ. The coefficient of friction between the block and the plane is μ and tanθ >μ. The block is held stationary by applying a force P parallel to the plane. The direction of force pointing up the plane is taken to be positive. As P is varied from $P_1 = mg(\sin\theta - \mu\cos\theta)$ to $P_2 = mg(\sin\theta + \mu\cos\theta)$, the frictional force f versus P graph will look like**.**

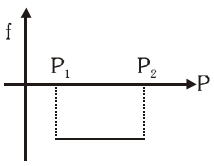
[IIT-JEE-2010]



(A) 

(B) 

(C) 

(D) 

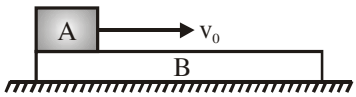
NL0117

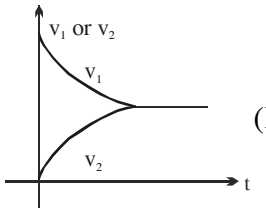
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\11-U & D, Basic Maths & Vector, NUM, K-11 D,2D\Eng\Eng\04_NUM & Friction\01_Eng.p65

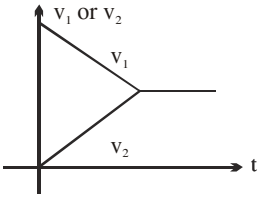
49. The rear side of a truck is open and a box of mass 20 kg is placed on the truck 4 m away from the open end, $\mu = 0.15$ and $g = 10\ m/s^2$. The truck starts from rest with an acceleration of $2m/s^2$ on a straight road. The distance moved by the truck when box starts fall down is :-

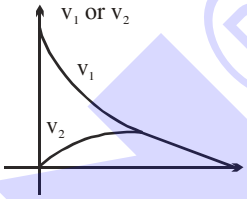
(A) 4 m (B) 8 m (C) 16 m (D) 32 m **NL0088**

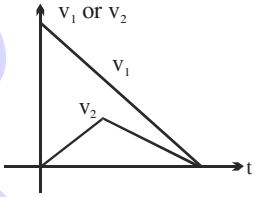
50. A block *A* is placed over a long rough plank *B* of same mass as shown in figure. The plank is placed over a smooth horizontal surface. At time t = 0, block *A* is given a velocity v_0 in horizontal direction. Let v_1 and v_2 be the velocities of *A* and *B* at time *t*. Then choose the correct graph between v_1 or v_2 and *t*.



(A) 

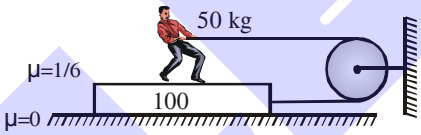
(B) 

(C) 

(D) 

NL0090

51. A man of mass 50 kg is pulling on a plank of mass 100 kg kept on a smooth floor as shown with force of 100 N. If both man & plank move together, find force of friction acting on man.



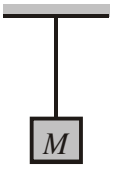
(A) $\frac{100}{3}$ N towards left (B) $\frac{100}{3}$ N towards right

(C) $\frac{250}{3}$ N towards left (D) $\frac{250}{3}$ N towards right **NL0092**

MULTIPLE CORRECT TYPE QUESTIONS

Newton's Laws of motion:

52. Consider a block suspended from a light string as shown in the figure. Which of the following pairs of forces constitute Newton's third law pair?



(A) Force with which string pulls on the ceiling and the force with which string pulls on block

(B) Force with which string pulls on the block and weight of the block

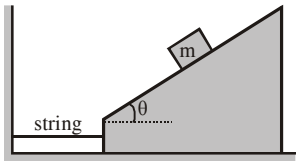
(C) Force acting on block due to the earth and force the block exerts on the earth

(D) Force with which block pulls on string and force with which the string pulls on the block

NL0096

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\1-U & D, Basic Maths & Vector, NLM\K-1 D,2D\Eng\Eng\04_NLM & Friction\01_Engp65

53. Refer the system shown in the figure. Block is sliding down the wedge. All surfaces are frictionless. Find correct statement(s)



(A) Acceleration of block is $g\sin\theta$ (B) Acceleration block is $g\cos\theta$

(C) Tension in the string is $mg\cos^2\theta$ (D) Tension in the string is $mg\sin\theta.\cos\theta$ **NL0073**

54. A carpenter of mass 50 kg is standing on a weighing machine placed in a lift of mass 20 kg. A light string is attached to the lift. The string passes over a smooth pulley and the other end is held by the carpenter as shown. When carpenter keeps the lift moving upward with constant velocity :- (g = 10 m/s^2)



(A) the reading of weighing machine is 15 kg

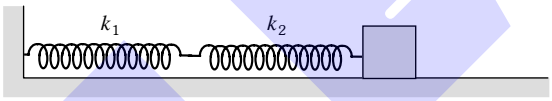
(B) the man applies a force of 350 N on the string

(C) net force on the man is 150 N

(D) Net force on the weighing machine is 150 N

NL0077

55. The mass in the figure can slide on a frictionless surface. When the mass is pulled out, spring 1 is stretched a distance x_1 and spring 2 is stretched a distance x_2. The spring constants are k_1 and k_2 respectively. Magnitude of spring force pulling back on the mass is



(A) k_1x_1 (B) k_2x_2 (C) $(k_1x_1 + k_2x_2)$ (D) $0.5\,(k_1 + k_2)\,(x_1 + x_2)$

NL0076

Friction:

56. A block is kept on a rough surface and applied with a horizontal force as shown which is gradually increasing from zero. The coefficient of static and kinetic friction are $1/\sqrt{3}$ then



(A) When F is less than the limiting friction, angle made by net force on the block by the surface is less than 30° with vertical.

(B) When the block is just about to move, the angle made by net force by the surface on the block becomes equal to 30° with vertical.

(C) When the block starts to accelerate, the angle made by net force by the surface on the block becomes constant and equal to 30° vertical.

(D) The angle made by net force with vertical on the block by the surface, depends on the mass of the block.

NL0101

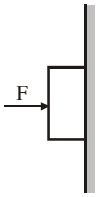
57. A block is placed over a plank. The coefficient of friction between the block and the plank is $\mu = 0.2$. Initially both are at rest, suddenly the plank starts moving with acceleration $a_0 = 4$ m/s^2. The displacement of the block in 1s is (g=10 m/s^2)

(A) 1 m relative to ground (B) 1 m relative to plank

(C) zero relative to plank (D) 2 m relative to ground **NL0103**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\11-U & D, Basic Maths & Vector, NLM, K-1 D, 2D\Eng\Eng\04_NLM & Friction\01_Eng.p65

58. A block of mass 1 kg is held at rest against a rough vertical surface by pushing by a force F horizontally. The coefficient of friction is 0.5. When

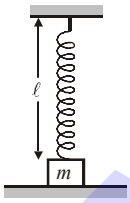


(A) F = 40 N, friction on the block is 20 N.
(B) F = 30 N, friction on the block is 10 N.
(C) F = 20 N, friction on the block is 10 N.
(D) Minimum value of force F to keep block at rest is 20 N. **NL0074**

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 59 to 61

A block of mass m is placed on a smooth horizontal floor is attached to one end of spring. The other end of the spring is attached to fixed support. When spring is vertical it is relaxed. Now the block is pulled towards right by a force F, which is being increased gradually. When the spring makes angle 53° with the vertical, block leaves the floor.



59. When blocks leaves the table, the normal force on it from table is

(A) mg (B) zero (C) $\frac{4mg}{3}$ (D) $\frac{3mg}{4}$ **NL0106**

60. Force constant of the spring is :-

(A) $\frac{5mg}{2\ell}$ (B) $\frac{15mg}{8\ell}$ (C) $\frac{5mg}{3\ell}$ (D) $\frac{5mg}{4\ell}$ **NL0106**

61. When the block leaves the table, the force F is :-

(A) $\frac{3mg}{4}$ (B) $\frac{4mg}{3}$ (C) $\frac{3mg}{5}$ (D) $\frac{4mg}{5}$ **NL0106**

Paragraph for Question No. 62 to 66

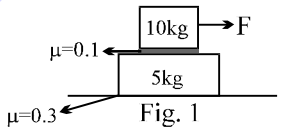


Fig. 1

62. When F = 2N, the frictional force between 5 kg block and ground is
(A) 2N (B) 0 (C) 8 N (D) 10 N **NL0107**

63. When F = 2N, the frictional force between 10 kg block and 5 kg block is
(A) 2N (B) 15 N (C) 10 N (D) None **NL0107**

64. The maximum F which will not cause motion of any of the blocks is
(A) 10 N (B) 15 N (C) data insufficient (D) None **NL0107**

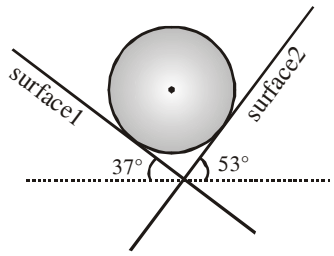
65. The maximum acceleration of 5 kg block is :-
(A) 1 m/s^2 (B) 3 m/s^2 (C) 0 (D) None **NL0107**

66. The acceleration of 10 kg block when F = 30 N is
(A) 2 m/s^2 (B) 3 m/s^2 (C) 1 m/s^2 (D) None **NL0107**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture \Phy\Module\11-U & D, Basic Maths & Vector, NUM, K-1 D,2D\Eng\Eng\04_NUM & Friction\01_Engp05

MATRIX MATCH TYPE QUESTION

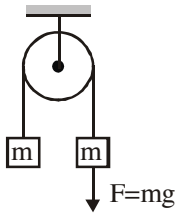
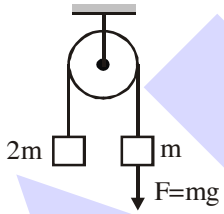
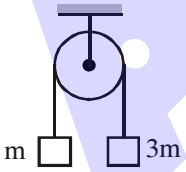
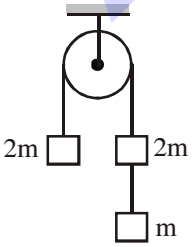
67. A sphere of mass 10 kg is placed in equilibrium in a V shaped groove plane made of two smooth surfaces 1 and 2 as shown in figure. ($g = 10\ ms^{-2}$)



	Column I		**Column II**
(A)	Normal reaction by Surface 1	(P)	Zero
(B)	Normal reaction by surface 2	(Q)	60 N
(C)	Force on sphere by Earth	(R)	80 N
(D)	Net force on sphere	(S)	100 N
		(T)	120 N

NL0084

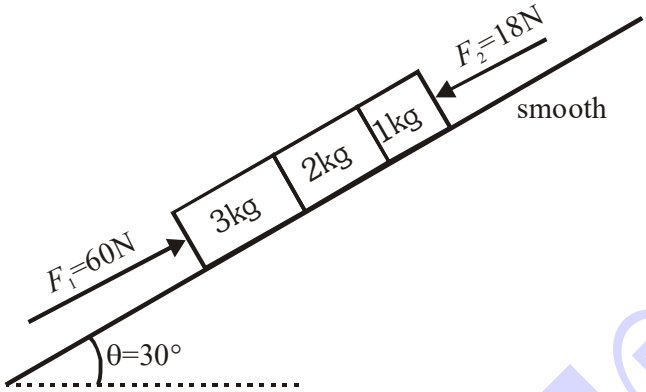
68. Match the situations in column I to the accelerations of blocks in the column II (acceleration due to gravity is g and F is an additional force applied to one of the blocks ?

	Column I		**Column II**
(A)		(P)	$\frac{g}{5}$
(B)		(Q)	$\frac{g}{3}$
(C)		(R)	$\frac{g}{2}$
(D)		(S)	$\frac{2g}{3}$
		(T)	zero

NL0083

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\11-U & D, Basic Maths & Vector, NLM,K-11D,2D\Eng\Eng\04_NLM & Friction\01_Eng.p65

69. In the diagram shown in figure (g = 10 m/s^2)



	Column I		**Column II**
(A)	Acceleration of 2 kg block in m/s^2	(P)	8
(B)	Net force on 3 kg block in newton	(Q)	25
(C)	Normal reaction between 2 kg and 1 kg in newton	(R)	2
(D)	Normal reaction between 3 kg and 2 kg in newton	(S)	45
		(T)	None

NL0082

ALLEN®

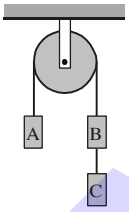
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\11-U & D, Basic Maths & Vector, NUM\K-11 D,2D\Eng\Eng\04_NUM & Friction\01_Engp65

EXERCISE (S)

Newton's Law of Motion:

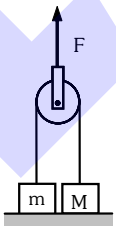
1. A force F applied to an object of mass m_1 produces an acceleration of 3.00 m/s^2. The same force applied to a second object of mass m_2 produces an acceleration of 1.00 m/s^2.
(i) What is the value of the ratio m_1/m_2 ?
(ii) If m_1 and m_2 are combined, find their acceleration under the action of the force F. **NL0001**

2. In the system shown, the blocks A, B and C are of weight 4W, W and W respectively. The system set free. The tension in the string connecting the blocks B and C is



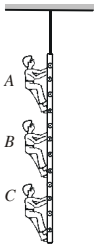
NL0002

3. In the system shown, pulley and strings are ideal. The vertically upward pull F is being increased gradually, find magnitude of F and acceleration of the 5 kg block at the moment the 10 kg block leaves the floor.



NL0004

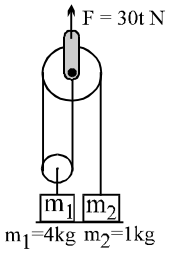
4. A ladder is hanging from ceiling as shown in figure. Three men A, B and C of masses 40 kg, 60 kg, and 50 kg are climbing the ladder. Man A is going up with retardation 2 m/s^2, C is going up with an acceleration of 1 m/s^2 and man B is going up with a constant speed of 0.5 m/s. Find the tension in the string supporting the ladder. [g = 9.8 m/s^2]



NL0027

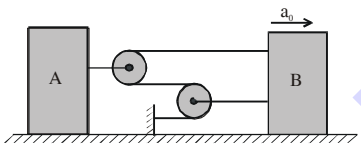
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\1-U & D, Basic Maths & Vector, NLM, K-1 D, 2D\Eng\Eng\04_NLM & Friction\01_Eng.p65

5. Force F is applied on upper pulley. If $F = 30t$ N where t is time in second. Find the time when m_1 loses contact with floor.



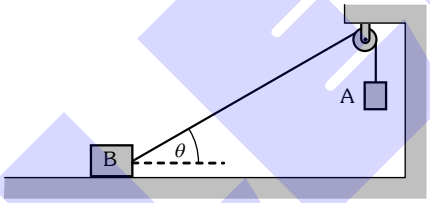
NL0005

6. Calculate the relative acceleration of A w.r.t. B if B is moving with acceleration a_0 towards right.



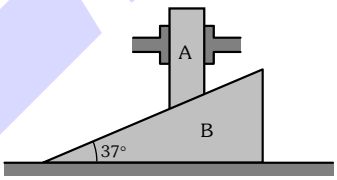
NL0015

7. The block A is moving downward with constant velocity v_0. Find the velocity of the block B, when the string makes an angle θ with the horizontal



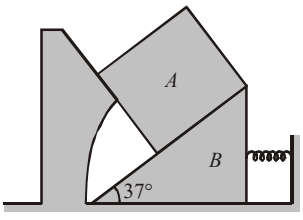
NL0012

8. Rod A can slide in vertical direction pushing the triangular wedge B towards right. The wedge is moving toward right with uniform acceleration a_B. Find acceleration of the rod A.



NL0014

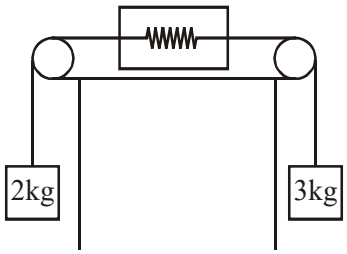
9. In the figure shown, all surfaces are smooth and block A and wedge B have mass 10 kg and 20 kg respectively. Find normal reaction between block A & B, spring force and normal reaction of ground on block B. (g= 10 m/s^2).



NL0008

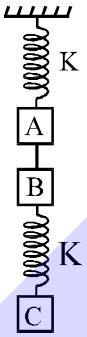
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\1-U & D, Basic Maths & Vector, NUM, K-1 D,2D\Eng\Eng\04_NUM & Friction\01_Engp65

10. Find the reading of the massless spring balance in the given condition



NL0009

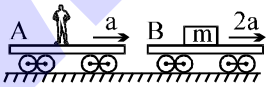
11. The system shown adjacent is in equilibrium. Find the acceleration of the blocks *A, B* & *C* all of equal masses *m* at the instant when: (Assume springs to be ideal)



(i) The spring between ceiling & *A* is cut.
(ii) The string (inextensible) between *A* & *B* is cut.
(iii) The spring between *B* & *C* is cut.
Also find the tension in the string when the system is at rest and in the above 3 cases.

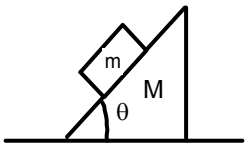
NL0010

12. Two trolley A and B are moving with accelerations a and 2a respectively in the same direction. To an observer in trolley A, the magnitude of pseudo force acting on a block of mass m on the trolley B is



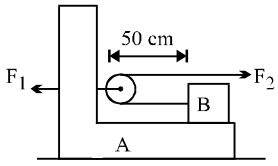
NL0018

13. A block of mass *m* lies on wedge of mass *M* as shown in figure.



With what minimum acceleration must the wedge be moved towards right horizontally so that block m falls freely. **NL0011**

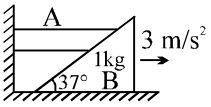
14. A 1kg block *B* rests as shown on a bracket *A* of same mass. Constant forces F_1 = 20N and F_2 = 8N start to act at time t = 0 when the distance of block *B* from pulley is 50cm. Time when block *B* reaches the pulley is _____.



NL0007

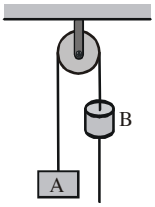
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\11-U & D, Basic Maths & Vector, NLM, K-11 D, 2D\Eng\Eng\04_NLM & Friction\01_Eng.p65

15. Find force in newton which mass *A* exerts on mass *B* if *B* is moving towards right with 3 m/s^2. Also find mass of *A*. (All surfaces are smooth)



NL0013

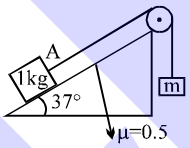
16. A 2 kg block *A* is attached to one end of a light string that passes over an an ideal pulley and a 1 kg sleeve *B* slides down the other part of the string with an acceleration of 5 m/s^2 with respect to the string. Find the acceleration of the block, acceleration of sleeve and tension in the string. [g = 10 m/s^2]



NL0032

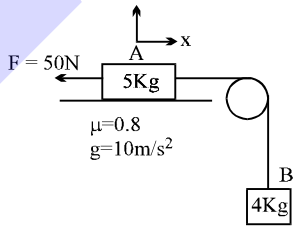
Friction:

17. In the figure, what should be mass m so that block *A* slides up with a constant velocity?



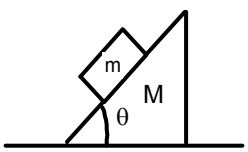
NL0023

18. Find the acceleration of the blocks and magnitude & direction of frictional force between block *A* and table, if block *A* is pulled towards left with a force of 50 N.



NL0024

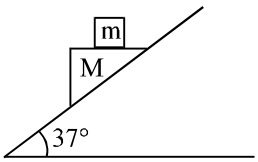
19. A block of mass *m* lies on wedge of mass *M* as shown in figure. Find the minimum friction coefficient required between wedge *M* and ground so that it does not move while block *m* slips down on it.



NL0021

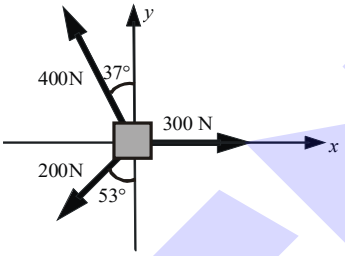
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture \Phy\Module\1-U & D, Basic Maths & Vector, NLM\K-1 D,2D\Eng\Eng\04_NLM & Friction\01_Engp65

20. Block *M* slides down on frictionless incline as shown. Find the minimum friction coefficient so that *m* does not slide with respect to *M*.



NL0025

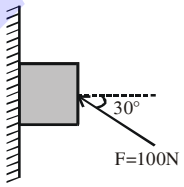
21. A block is placed on a rough horizontal plane. Three horizontal forces are applied on the block as shown in the figure. If the block is in equilibrium, find the friction force acting on the block.



NL0016

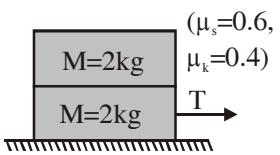
22. A thin rod of length 1 m is fixed in a vertical position inside a train, which is moving horizontally with constant acceleration 4 m/s^2. A bead can slide on the rod, and friction coefficient between them is 1/2. If the bead is released from rest at the top of the rod, find the time when it will reach at the bottom.[g = 10 m/s^2] **NL0019**

23. A force of 100 N is applied on a block of mass 3 kg as shown in figure. The coefficient of friction between the wall and the surface of the block is 1/4. Calculate frictional force acting on the block.



NL0017

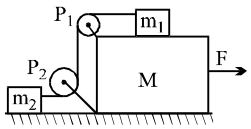
24. The coefficient of static and kinetic friction between the two blocks and also between the lower block and the ground are μ_S= 0.6 and μ_K = 0.4. Find the value of tension *T* applied on the lower block at which the upper block begins to slip relative to lower block.



NL0033

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\11-U & D, Basic Maths & Vector, NLM,K-11D,2D\Eng\Eng\04_NLM & Friction\01_Eng.p65

25. In the figure masses m_1, m_2 and M are 20 kg, 5 kg and 50 kg respectively. The co-efficient of friction between M and ground is zero. The co-efficient of friction between m_1 and M and that between m_2 and ground is 0.3. The pulleys and the string are massless . The string is perfectly horizontal between P_1 and m_1 and also between P_2 and m_2 . The string is perfectly vertical between P_1 and P_2.An external horizontal force F is applied to the mass M. Take g = 10 m/s^2. **[IIT-JEE 2000]**

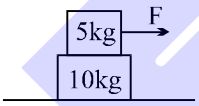


(i) Draw a free-body diagram for mass M, clearly showing all the forces.

(ii) Let the magnitude of the force of friction between m_1 and M be f_1 and that between m_2 and ground be f_2. For a particular F it is found that $f_1 = 2 f_2$. Find f_1 and f_2 . Write down equations of motion of all the masses . Find F, tension in the string and accelerations of the masses.

NL0034

26. Coefficient of friction between 5 kg and 10 kg block is 0.5. If friction between them is 20 N. What is the value of force being applied on 5 kg. The floor is frictionless.



NL0026

27. A block of mass 1 kg is horizontally thrown with a velocity of 10 m/s on a stationary long plank of mass 2 kg whose surface has $\mu = 0.5$. Plank rests on frictionless surface. Find the time when block comes to rest w.r.t. plank. **NL0020**

28. A block is moving on an inclined plane making an angle 45° with the horizontal and the coefficient of friction is μ. The force required to just push it up the inclined plane is 3 times the force required to just prevent it from sliding down. If we define $N = 10\mu$, then N is **[IIT-JEE-2011]**

NL0118

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\11-U & D, Basic Maths & Vector, NLM\K-11 D,2D\Eng\Eng\04_NLM & Friction\01_Engp65

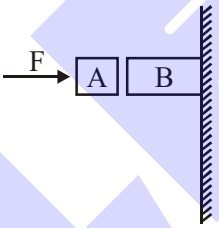
EXERCISE (JM)

1. A block of mass m is placed on a surface with a vertical cross section given by $y = \frac{x^3}{6}$. If the coefficient of friction is 0.5, the maximum height above the ground at which the block can be placed without slipping is :- **[JEE-Main-2014]**

(1) $\frac{1}{3}m$ (2) $\frac{1}{2}m$ (3) $\frac{1}{6}m$ (4) $\frac{2}{3}m$

NL0112

2. Given in the figure are two blocks A and B of weight 20 N and 100 N, respectively. These are being pressed against a wall by a force F as shown. If the coefficient of friction between the blocks is 0.1 and between block B and the wall is 0.15, the frictional force applied by the wall on block B is :- **[JEE-Main-2015]**

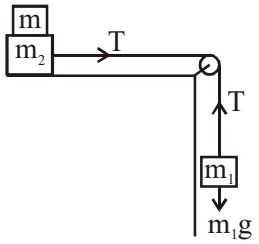


(1) 120 N (2) 150 N (3) 100 N (4) 80 N **NL0113**

3. A rocket is fired vertically from the earth with an acceleration of 2g, where g is the gravitational acceleration. On an inclined plane inside the rocket, making an angle θ with the horizontal, a point object of mass m is kept. The minimum coefficient of friction μ_{min} between the mass and the inclined surface such that the mass does not move is: **[JEE-Main Online-2016]**

(1) 2 tan θ (2) 3 tan θ (3) tan θ (4) tan 2θ **NL0114**

4. Two masses m_1 = 5kg and m_2 = 10kg, connected by an inextensible string over a frictionless pulley, are moving as shown in the figure. The coefficient of friction of horizontal surface is 0.15. The minimum weight m that should be put on top of m_2 to stop the motion is :- **[JEE-Main-2018]**



(1) 27.3 kg (2) 43.3 kg (3) 10.3 kg (4) 18.3 kg

NL0115

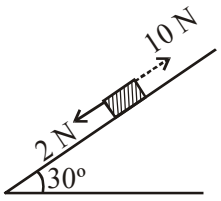
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\11-U & D, Basic Maths & Vector, NLM, K-11\2D\Eng\Eng\04_NLM & Friction\01_Eng.p65

SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

5. A mass of 10 kg is suspended vertically by a rope from the roof. When a horizontal force is applied on the rope at some point, the rope deviated at an angle of 45° at the roof point. If the suspended mass is at equilibrium, the magnitude of the force applied is ($g = 10\ ms^{-2}$) **[JEE-Main-2019_Jan]**

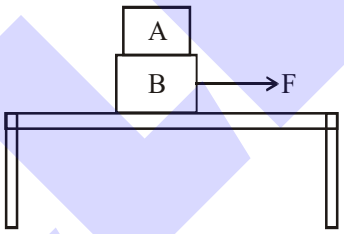
(1) 200 N (2) 100 N (3) 140 N (4) 70 N **NL0120**

6. A block kept on a rough inclined plane, as shown in the figure, remaias at rest upto a maximum force 2 N down the inclined plane. The maximum external force up the inclined plane that does not move the block is 10 N. The coefficient of static friction betwreen the block and the plane is : [Take $g = 10\ m/s^2$] **[JEE-Main-2019_Jan]**



(1) $\frac{2}{3}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{\sqrt{3}}{4}$ (4) $\frac{1}{2}$ **NL0123**

7. Two blocks A and B of masses $m_A = 1$ kg and $m_B = 3$ kg are kept on the table as shown in figure. The coefficient of friction between A and B is 0.2 and between B and the surface of the table is also 0.2. The maximum force F that can be applied on B horizontally, so that the block A does not slide over the block B is : (Take $g = 10\ m/s^2$) **[JEE-Main-2019_April]**

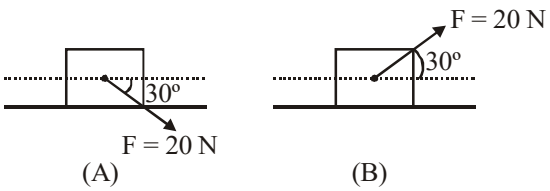


(1) 16 N (2) 40 N (3) 12 N (4) 8 N **NL0124**

8. A spring whose unstretched length is l has a force constant k. The spring is cut into two pieces of unstretched lengths l_1 and l_2 where, $l_1 = nl_2$ and n is an integer. The ratio k_1/k_2 of the corresponding force constants, k_1 and k_2 will be : **[JEE-Main-2019_April]**

(1) $\frac{1}{n^2}$ (2) n^2 (3) $\frac{1}{n}$ (4) n **NL0125**

9. A block of mass 5 kg is (i) pushed in case (A) and (ii) pulled in case (B), by a force F = 20 N, making an angle of 30° with the horizontal, as shown in the figures. The coefficient of friction between the block and floor is $\mu = 0.2$. The difference between the accelerations of the block, in case (B) and case (A) will be : ($g = 10\ ms^{-2}$) **[JEE-Main-2019_April]**



(1) 0 ms^{-2} (2) 0.8 ms^{-2} (3) 0.4 ms^{-2} (4) 3.2 ms^{-2} **NL0126**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\1-U & D, Basic Maths & Vector, NUM, K-1 D,2D\Eng\Eng\04_NUM & Friction\01_Engp05

10. A ball is thrown upward with an initial velocity V_0 from the surface of the earth. The motion of the ball is affected by a drag force equal to $m\gamma\upsilon^2$ (where m is mass of the ball, υ is its instantaneous velocity and γ is a constant). Time taken by the ball to rise to its zenith is :

[JEE-Main-April-2019]

(1) $\frac{1}{\sqrt{\gamma g}}\sin^{-1}\left(\sqrt{\frac{\gamma}{g}}V_0\right)$

(2) $\frac{1}{\sqrt{\gamma g}}\tan^{-1}\left(\sqrt{\frac{\gamma}{g}}V_0\right)$

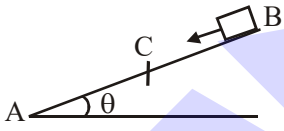
(3) $\frac{1}{\sqrt{2\gamma g}}\tan^{-1}\left(\sqrt{\frac{2\gamma}{g}}V_0\right)$

(4) $\frac{1}{\sqrt{\gamma g}}\ln\left(1+\sqrt{\frac{\gamma}{g}}V_0\right)$

NL0156

11. A small block starts slipping down from a point B on an inclined plane AB, which is making an angle θ with the horizontal section BC is smooth and the remaining section CA is rough with a coefficient of friction μ. It is found that the block comes to rest as it reaches the bottom (point A) of the inclined plane. If BC = 2AC, the coefficient of friction is given by $\mu = k\tan\theta$). The value of k is _____.

[JEE-Main-2020_Sept]



NL0157

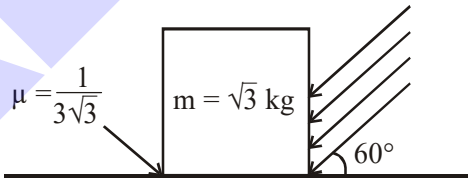
12. The coefficient of static friction between a wooden block of mass 0.5 kg and a vertical rough wall is 0.2. The magnitude of horizontal force that should be applied on the block to keep it adhere to the wall will be _____N. [$g = 10\ ms^{-2}$] **[JEE-Main-2021_Feb]**

NL0158

13. As shown in the figure, a block of mass $\sqrt{3}$ kg is kept on a horizontal rough surface of coefficient of friction $\frac{1}{3\sqrt{3}}$. The critical force to be applied on the vertical surface as shown at an angle 60° with horizontal such that it does not move, will be 3x. The value of x will be

[$g = 10\ m/s^2$; $\sin 60° = \frac{\sqrt{3}}{2}$; $\cos 60° = \frac{1}{2}$]

[JEE-Main-2021_Feb]



NL0159

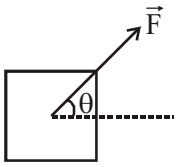
14. A block of mass m slides along a floor while a force of magnitude F is applied to it at an angle θ as shown in figure. The coefficient of kinetic friction is μ_K. Then, the block's acceleration 'a' is given by : (g is acceleration due to gravity) **[JEE-Main-2021_March]**

(1) $-\frac{F}{m}\cos\theta-\mu_K\left(g-\frac{F}{m}\sin\theta\right)$

(2) $\frac{F}{m}\cos\theta-\mu_K\left(g-\frac{F}{m}\sin\theta\right)$

(3) $\frac{F}{m}\cos\theta-\mu_K\left(g+\frac{F}{m}\sin\theta\right)$

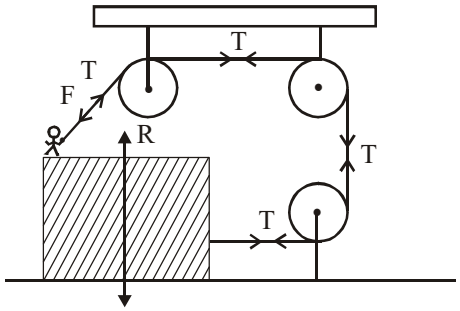
(4) $\frac{F}{m}\cos\theta+\mu_K\left(g-\frac{F}{m}\sin\theta\right)$



NL0160

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\1-U & D, Basic Maths & Vector, NLM, K-1 D, 2D\Eng\Eng\04_NLM & Friction\01_Eng.p65

15. A boy of mass 4 kg is standing on a piece of wood having mass 5kg . If the coefficient of friction between the wood and the floor is 0.5, the maximum force that the boy can exert on the rope so that the piece of wood does not move from its place is _____N.(Round off to the Nearest Integer) [Take g = 10 ms^{-2}] **[JEE-Main-2021_March]**



NL0161

16. A particle of mass M originally at rest is subjected to a force whose direction is constant but magnitude varies with time according to the relation

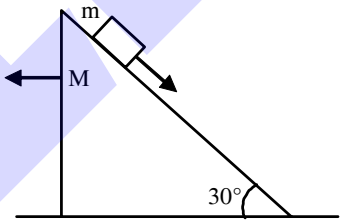
$$F = F_0\left[1-\left(\frac{t-T}{T}\right)^2\right]$$

Where F_0 and T are constants. The force acts only for the time interval 2T. The velocity v of the particle after time 2T is : **[JEE-Main-2021_July]**

(1) $2F_0T / M$ (2) $F_0T / 2M$ (3) $4F_0T / 3M$ (4) $F_0T / 3M$

NL0162

17. A block of mass m slides on the wooden wedge, which in turn slides backward on the horizontal surface. The acceleration of the block with respect to the wedge is : Given m = 8 kg, M = 16 kg Assume all the surfaces shown in the figure to be frictionless. **[JEE-Main-2021_Aug]**



(1) $\frac{4}{3}g$ (2) $\frac{6}{5}g$ (3) $\frac{3}{5}g$ (4) $\frac{2}{3}g$

NL0163

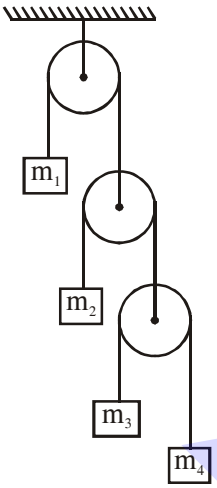
18. When a body slides down from rest along a smooth inclined plane making an angle of 30° with the horizontal, it takes time T. When the same body slides down from the rest along a rough inclined plane making the same angle and through the same distance, it takes time αT, where is α constant greater than 1. The co-efficient of friction between the body and the rough plane is $\frac{1}{\sqrt{x}}\left(\frac{\alpha^2-1}{\alpha^2}\right)$ where x = ... **[JEE-Main-2021_Aug]**

NL0164

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture \Phy\Module\1-U & D, Basic Maths & Vector, NUM, K-1 D,2D\Eng\Eng\04_NUM & Friction\01_Engp65

19. In the arrangement shown in figure a_1, a_2, a_3 and a_4 are the accelerations of masses m_1, m_2, m_3 and m_4 respectively. Which of the following relation is true for this arrangement?

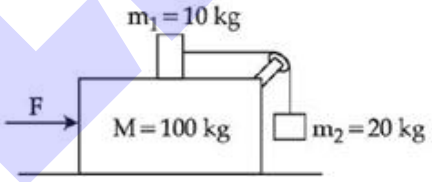
[JEE-Main-2022_June]



(1) $4a_1 + 2a_2 + a_3 + a_4 = 0$
(2) $a_1 + 4a_2 + 3a_3 + a_4 = 0$
(3) $a_1 + 4a_2 + 3a_3 + 2a_4 = 0$
(4) $2a_1 + 2a_2 + 3a_3 + a_4 = 0$

NL0165

20. Three masses M = 100 kg, m_1 = 10 kg and m_2 = 20 kg are arranged in a system as shown in figure. All the surfaces are frictionless and strings are inextensible and weightless. The pulleys are also weightless and frictionless. A force F is applied on the system so that the mass m_2 moves upward with an acceleration of 2 ms^{-2}. The value of F is : (Take g = 10 ms^{-2}) **[JEE-Main-2022_July]**



(1) 3360 N
(2) 3380 N
(3) 3120 N
(4) 3240 N

NL0166

21. A bag is gently dropped on a conveyor belt moving at a speed of 2 m/s. The coefficient of friction between the conveyor belt and bag is 0.4 Initially, the bag slips on the belt before it stops due to friction. The distance travelled by the bag on the belt during slipping motion is : [Take g = 10 m/s^{-2}]

[JEE-Main-2022_July]

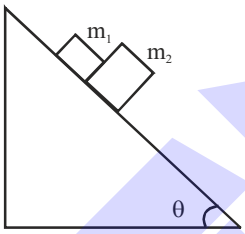
(1) 2 m
(2) 0.5 m
(3) 3.2 m
(4) 0.8 ms

NL0167

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\11-U & D, Basic Maths & Vector, NLM, K-11D,2D\Eng\Eng\04_NLM & Friction\01_Eng.p65

EXERCISE (JA)

1. A block of mass m_1 = 1 kg another mass m_2 = 2kg, are placed together (see figure) on an inclined plane with angle of inclination θ. Various values of θ are given in List I. The coefficient of friction between the block m_1 and the plane is always zero. The coefficient of static and dynamic friction between the block m_2 and the plane are equal to $\mu = 0.3$. In List II expressions for the friction on block m_2 are given. Match the correct expression of the friction in List II with the angles given in List I, and choose the correct option. The acceleration due to gravity is denoted by g.
[**useful information :** $\tan(5.5°) \approx 0.1$; $\tan(11.5°) \approx 0.2$; $\tan(16.5°) \approx 0.3$] **[IIT-JEE-2014]**



	List–I		**List–II**
(P)	$\theta = 5°$	(1)	$m_2 g \sin\theta$
(Q)	$\theta = 10°$	(2)	$(m_1 + m_2)g \sin\theta$
(R)	$\theta = 15°$	(3)	$\mu m_2 g \cos\theta$
(S)	$\theta = 20°$	(4)	$\mu(m_1 + m_2)g \cos\theta$

Code :

(A) P-1, Q-1, R-1, S-3
(B) P-2. Q-2, R-2, S-3
(C) P-2,. Q-2, R-2, S-4
(D) P-2, Q-2, R-3, S-3

NL0119

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\11-U & D, Basic Maths & Vector, NUM, K-11 D,2D\Eng\Eng\04_NUM & Friction\01_Engp05

ANSWER KEY

EXERCISE # O

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	C	D	D	C	C	D	B	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	C	C	B	A	C	B	B	A	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	A	B	D	C	D	B	C	A	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	C	B	C	B	B	B	C	A	C
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	B	D	B	A	B	A	A	A	C	B
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	A	C,D	A,D	A,B	A,B	A,B,C,D	A,B	B,C,D	B	A
Que.	61	62	63	64	65	66	67	68	69	
Ans.	B	A	A	A	C	A	A-R; B-Q; C-S; D-P	A-R; B-T; C-R; D-P	A-R; B-T; C-Q; D-T	

EXERCISE # S

1. (i) $\frac{m_1}{m_2}=\frac{1}{3}$ (ii) $a = 3/4$ m/s^2 **2.** $\frac{4}{3}W$ **3.** 200 N, 10 m/s^2 **4.** 1440 N **5.** 2 sec

6. $\frac{a_0}{2}$ **7.** $v_0/\cos\theta$ **8.** $3a_B/4$ **9.** 80 *N*, 48 *N*, 264 *N* **10. 24N**

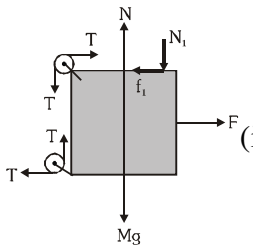
11. (i) $a_A=\frac{3g\downarrow}{2}=a_B$; $a_C=0$; $T = mg/2$; (ii) $a_A=2g\uparrow$, $a_B=2g\downarrow$, $a_c = 0$, $T = 0$;

(iii) $a_A=a_B=g/2\uparrow$, $a_c = g\downarrow$, $T=\frac{3mg}{2}$ **12. (ma)** **13.** $a = g\cot\theta$ **14.** 0.5 s

15. 5N, 16/31 kg **16.** 5 m/s^2 downwards, 0 m/s^2, 10 N **17.** 1 kg

18. $10\hat{i}$ **19.** $\mu_{min}=\frac{m\sin\theta\cos\theta}{m\cos^2\theta+M}$ **20.** 3/4 **21.** $\left(100\hat{i}-200\hat{j}\right)$ N

22. 1/2 s **23.** 20 N vertically downward **24**. 40 N

25. (i) 

(ii) a = 3/5 m/s^2 , T = 18 N, F = 60N **26.** 30 N **27.** 4/3 s **28. 5**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\11-U & D, Basic Maths & Vector, NLM,K-11D,2D\Eng\Eng\04_NLM & Friction\01_Eng.p65

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	1	3	1	2	2	1	3	2	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	25	3	2	30	3	4	3	1	NTA : 3 Allen : 1
Que.	21									
Ans.	2									

EXERCISE # JEE-ADVANCED

Que.	1
Ans.	D

ALLEN®

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\11-U & D, Basic Maths & Vector, NLM\K-11 D,2D\Eng\Eng\04_NLM & Friction\01_Engp65